Arij Asad

Doctoral Researcher in Applied Mathematics and Fluid Dynamics

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Research Profile

Doctoral researcher in applied mathematics and fluid dynamics at the University of Bristol, studying buoyancy-driven mixing and plume-generated stratification in confined environments, with applications to the storage of nuclear waste. Combines planar laser-induced fluorescence (PLIF) experiments with similarity analysis, mathematical modelling and finite-element simulations of confined and gravity-driven flows.

Research interests: buoyant plumes; mixing and stratification; experimental fluid dynamics; confined and gravity-driven flows; computational fluid dynamics; finite-element methods.

Education

Doctor of Philosophy (PhD); research in applied mathematics and fluid dynamics

School of Electrical, Electronic and Mechanical Engineering, University of Bristol | Nov 2022–present

- Thesis: *Mixing behaviour of plumes*.
- Supervisors: Dr Andrew Lawrie and Dr Stuart Thomson.
- Expected thesis submission: 30 October 2026.
- Research funded by Nuclear Waste Services (NWS) through its Research Support Office (RSO).

BA (Cantab), Mathematics (Mathematical Tripos)

Murray Edwards College, University of Cambridge | Oct 2019–Jun 2022

- MA (Cantab), conferred Apr 2026.

Research Experience

Doctoral Researcher

University of Bristol | Nov 2022–present

- Investigate mixing and stratification generated by laminar, transitional and turbulent buoyant plumes in confined rectangular and circular geometries, with and without laminar crossflow, motivated by hydrogen ventilation in geological disposal facilities.
- Designed and conducted PLIF experiments to obtain time-resolved concentration and density fields for single-source, twin-source and ventilated configurations.
- Designed water-tunnel and crossflow components using Fusion 360 and CAD, working with mechanical workshops to refine geometry for manufacture and assembly.
- Developed scaling laws and similarity solutions for stratified-layer growth and profile evolution, including the effects of plume regime, crossflow and enclosure geometry.
- Built finite-element simulations in FEniCS for incompressible flow and advection–diffusion and ran computational studies on Bristol's Slurm high-performance computing systems.

MSc Research Project Co-supervisor

University of Bristol | Jan 2026–Sep 2026

Co-supervised an MSc research project entitled *Stratification Development in a Confined Region: An Experimental Comparison of Density-Layer Development Generated by Single- and Double-Nozzle Buoyant Flows*, providing day-to-day research guidance.

Cambridge Mathematics Open Intern

Faculty of Mathematics, University of Cambridge | Jul–Aug 2022

Completed an eight-week research project on a Navier-slip ellipsoid in shear flow, supervised by Dr Catherine Kamal; presented the work at the Cambridge Mathematics Placements Presentation Day.

Manuscripts, Research Software and Datasets

Manuscripts in preparation

- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *On the mixing behaviour of plumes in a confined environment*.
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a laminar crossflow*.
- **Arij Asad**, Andrew G. W. Lawrie and Stuart J. Thomson, *Stratification development in a confined circular domain with and without laminar crossflow*.

Research software

- **PLIF acquisition and control software:** C client–server tools for synchronising laser pulsing and DALSA camera image capture across a host computer and BeagleBone Black. Archived release (2026).
- **MATLAB PLIF analysis workflow:** fluorescence calibration and image processing, density-field reconstruction and stratification analysis. Archived release (2026).
- **FEniCS stratification models:** finite-element models of flow and stratification in Cartesian, axisymmetric and three-dimensional rectangular geometries, with verification, mesh and time-step sensitivity analysis and post-processing tools. Archived release (2026).

Research datasets (Zenodo)

- **Arij Asad** (2026). *Experimental plume apparatus: intake design, CAD models and supporting calculations*.
- **Arij Asad** (2026). *2D Cartesian plume simulations: fields, profiles and resolution comparisons*.
- **Arij Asad** (2026). *2D axisymmetric plume simulations: fields, profiles and resolution comparisons*.
- **Arij Asad** (2026). *3D rectangular plume simulations: fields, diagnostics and mesh-sensitivity comparisons*.

Talks and Presentations

Osborne Reynolds Day Jun 2026
On the mixing behaviour of plumes in a confined environment (poster)

British Applied Mathematics Colloquium (BAMC), Norwich Mar 2026
On the mixing behaviour of plumes in a confined environment (poster)

Nuclear Waste Services Research Support Office Annual Conference, Manchester Jan 2026
How does hydrogen build up in a closed vault environment? (poster)

2nd European Fluid Dynamics Conference, Dublin Aug 2025
Mixing behaviour of plumes

NWS RSO Annual Conference, Bristol Jan 2025
Hydrogen ventilation in a GDF (poster)

NWS RSO Annual Conference, Sheffield Jan 2024
Mixing behaviour of laminar plumes with and without a ventilating crossflow (invited technical talk)

NWS RSO Advanced Manufacturing Webinar (online) Oct 2023
Ventilation of hydrogen in UILW vaults (recording)

NWS RSO Annual Conference, Leeds Jan 2023
Hydrogen ventilation in a geological disposal facility

Cambridge Mathematics Placements Presentation Day Aug 2022
The motion of anisotropic carbon nanoparticles in shear flow: a hydrodynamic perspective

Murray Edwards College Maths Circle, Cambridge 2019
Sphere packing (talk for the Murray Edwards undergraduate cohort)

Technical Skills

Experimental methods: PLIF acquisition and calibration; MATLAB image processing; extraction and analysis of time-resolved concentration and density fields.

Apparatus design: Fusion 360 and CAD; experimental-rig development and coordination with mechanical and electronics workshops.

Programming and computing: MATLAB, Python, C, Git/GitHub, Linux/Bash and LaTeX; Slurm HPC on Bristol's BlueCrystal and BluePebble systems.

Numerical methods: Scaling and similarity analysis; FEniCS finite-element simulation of incompressible flow and advection–diffusion.

Languages: Urdu (reading and conversational proficiency).

Teaching and Supervision

Undergraduate Mathematics Supervisor

University of Cambridge colleges | Oct 2024–present

- Appointed to deliver Part II Fluid Dynamics II supervisions to 44 students across 14 Cambridge colleges over the 2024/25–2026/27 academic years.
  Colleges: Churchill, Corpus Christi, Emmanuel, Fitzwilliam, Girton, Jesus, Lucy Cavendish, Magdalene, Murray Edwards, Newnham, Pembroke, Queens', St John's and Sidney Sussex.
- Mark written work and lead example-sheet and revision supervisions.
- Provide postgraduate references and academic progression support when requested.

Mathematics Tutor and Mock Interviewer

Dipont International | Oct 2025–present

- Provide admissions-test preparation, personal statement guidance and mock interviews for Oxford and Cambridge applicants.
- Delivered 33 hours of intensive small-group mathematics teaching and mock-interview preparation on a two-week Wuxi residential programme in Jul 2026, with structured feedback.

Graduate Teacher Level 1 and Demonstrator

University of Bristol | Sep 2023–Jan 2026

- Prepared and delivered weekly tutorials in Probability and Statistics and Introduction to Pure Mathematics as Graduate Teacher Level 1.
- Demonstrated Engineering Mathematics, Methods of Applied Mathematics, Applied Linear Algebra and Continuum Mathematics; provided engineering mathematics drop-in support.

Private Mathematics Tutor

Arij Asad Maths; primarily online, with occasional in-person teaching | Oct 2019–present

- Provide direct individual tuition in A-level Mathematics, Further Mathematics, Physics and Computer Science, together with undergraduate mathematics topics.
- Prepare candidates for STEP and TMUA (and MAT, 2020–2025) through diagnostic work, timed papers and Cambridge-style mock interviews.

Admissions Mathematics Tutor and Mock Interviewer

Blue Education; independent contractor | Nov 2019–present

Conduct Oxbridge Mathematics mock interviews and provide written feedback.

Academic Leadership

She Talks Science STEP Summer School

Murray Edwards College, University of Cambridge | Director, 2025; Deputy Director, 2023–2024; Tutor, 2019–2022

- As Director, led a four-day residential programme for 50 students; co-led the selection of 50 participants from 184 applicants; coordinated five tutors, three supervisors and three enrichment speakers.
- Led the academic programme, timetable, staff briefing and safeguarding; taught inequalities classes and Cambridge-style supervisions.
- As Deputy Director, contributed to teaching materials, classes, supervisions and residential activities.

Admissions, Examining and Olympiads

STEP Supervising Assessor
OCR (Oxford Cambridge and RSA) | Jun–Jul 2026

- Supervised marking of STEP 2 Mechanics questions 9 and 10, providing real-time guidance to markers and checking scripts for consistent application of the mark scheme.
- Sampled marked scripts, documented quality checks and reviewed scripts requiring manual allocation before distribution to marking teams.

STEP Examination Marker
OCR (Oxford Cambridge and RSA) | 2024–present

Marked STEP 3 in 2024–2026 and STEP 2 in 2025–2026, including Mechanics in 2026; applied mark schemes consistently and provided feedback on their implementation.

Mathematics Undergraduate Admissions Interviewer
Homerton College, University of Cambridge | Dec 2024

Interviewed and assessed Mathematics applicants during a four-day online admissions cycle as part of a two-person panel.

Assessment Associate (Question Vetting)
Pearson VUE | Oct–Nov 2024

Reviewed confidential mathematics assessment material for technical accuracy and clarity.

Undergraduate Mathematics Marker
University of Bristol | 2024–2025

Marked Applied Analysis B, Probability and Statistics, Applied Linear Algebra coursework, Engineering Mathematics and Introduction to Pure Mathematics.

UK Mathematical Olympiad for Girls Vetter and Marker
United Kingdom Mathematics Trust | 2019

Reviewed draft problems for mathematical correctness, scope, difficulty and clarity, and marked Olympiad scripts as part of the national marking team.

Outreach and Enrichment

Widening Participation Tutor
University of Bristol | Jul 2023–Oct 2026

- Delivered university-access sessions and research talks for school pupils aged 11–18 through Insight into Bristol, Next Step Bristol, Virtual Summer School and Sutton Trust Summer School programmes.
- Developed KS5 fluid-mechanics outreach materials, including a lesson plan, presentation slides and interactive Mentimeter activities.
- Delivered *Fluid Mechanics and Nuclear Power* for the May 2025 Virtual Summer School; the joint engineering session received positive participant feedback on engagement, clarity and learning.

Admissions Talks and Webinars
2020–2026

- COMPOS interview-preparation webinar (through Blue Education), Aug 2026.
- Dipont International Oxbridge admissions talk, Shanghai, Jul 2026.
- Blue Education Oxbridge Interview Seminar, Oct 2025.
- Mathematics Open Day, University of Cambridge, 2022.
- Sutton Trust Experience Cambridge Day, 2020–2022.

Summer School Teaching
University of Cambridge | 2019–2022

Taught on the Sutton Trust and Intro to Maths at Uni summer schools.